Bird Species Observation Analysis

📌 Introduction

This project focuses on analysing observations of bird species across forest and grassland ecosystems. The aim is to understand biodiversity patterns, environmental impacts, and species behaviour using real-world data.

🎯 Objectives

- Study species distribution across habitats
- Analyse temporal patterns of bird activity
- Evaluate environmental conditions affecting observations
- Identify biodiversity hotspots
- Build an interactive dashboard
- Develop predictive insights

🛠️ Methodology

1. Data Cleaning

- Removed duplicates
- Handled missing values using imputation
- Converted date into usable format
- Extracted the month for temporal analysis

2. Exploratory Data Analysis

- Species distribution
- Habitat comparison
- Monthly trends
- Observer analysis

3. Visualisation

- Bar charts for species distribution
- Line charts for trends
- Scatter plots for environmental impact
- Heatmaps and hotspot analysis

📊 Key Insights

1. Species Distribution

- Grasslands show higher observation density
- Forests support specialised species

2. Temporal Trends

- Bird activity varies across months
- Peak activity observed in specific seasons

3. Environmental Impact

- Moderate temperature increases activity
- Weather conditions affect visibility

4. Detection Behaviour

- Most birds were detected within short distances
- Singing is the most common detection method

💼 Business Use Cases

- Wildlife conservation planning
- Eco-tourism development
- Environmental policy making
- Habitat restoration strategies

🤖 Machine Learning

A Random Forest model was used to predict species occurrence based on:

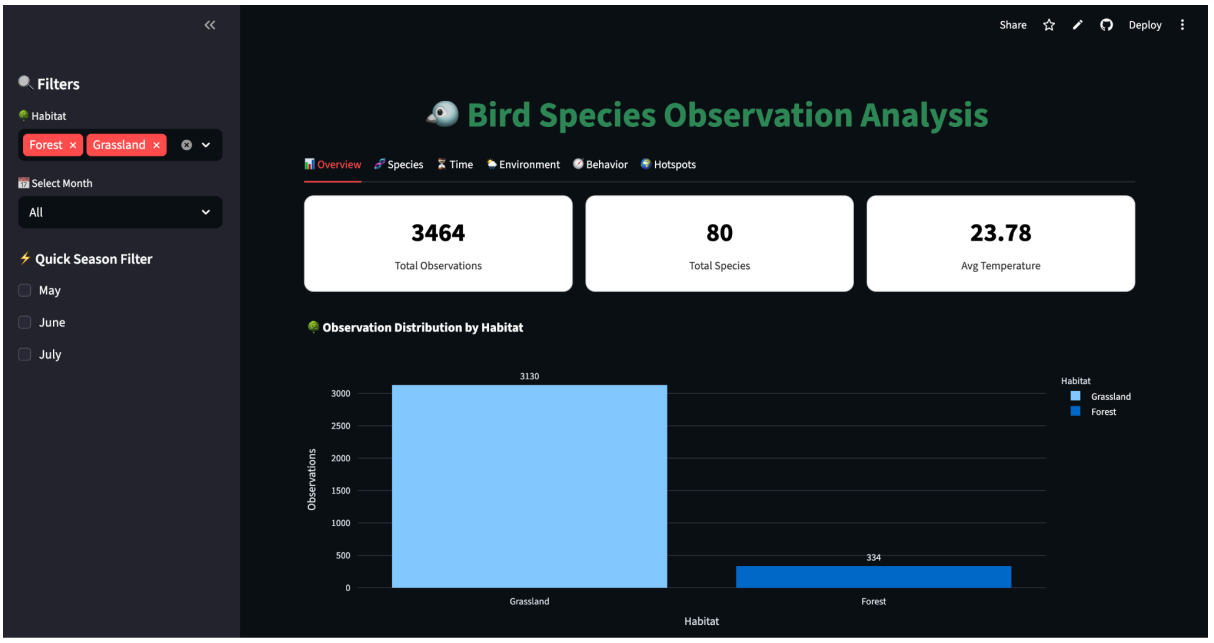
- Temperature
- Humidity
- Habitat

🚀 Conclusion

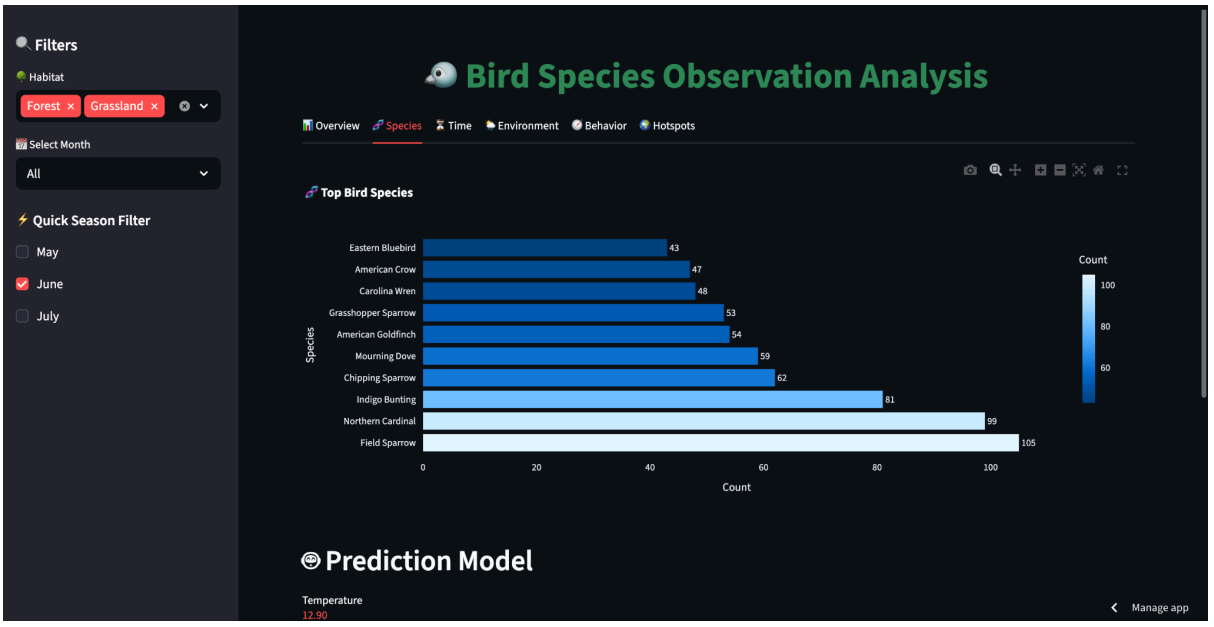
This project provides valuable insights into bird biodiversity and environmental relationships. The dashboard enables interactive exploration and supports data-driven decision-making.

📸 Screenshots List

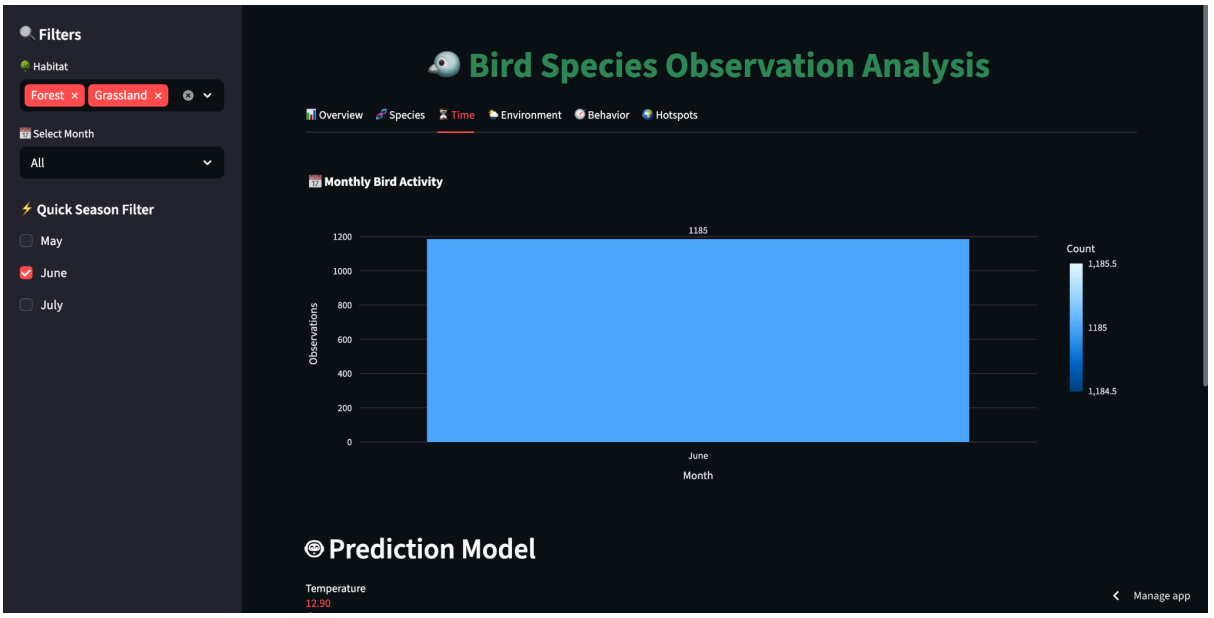
1. Dashboard Overview (KPIs)



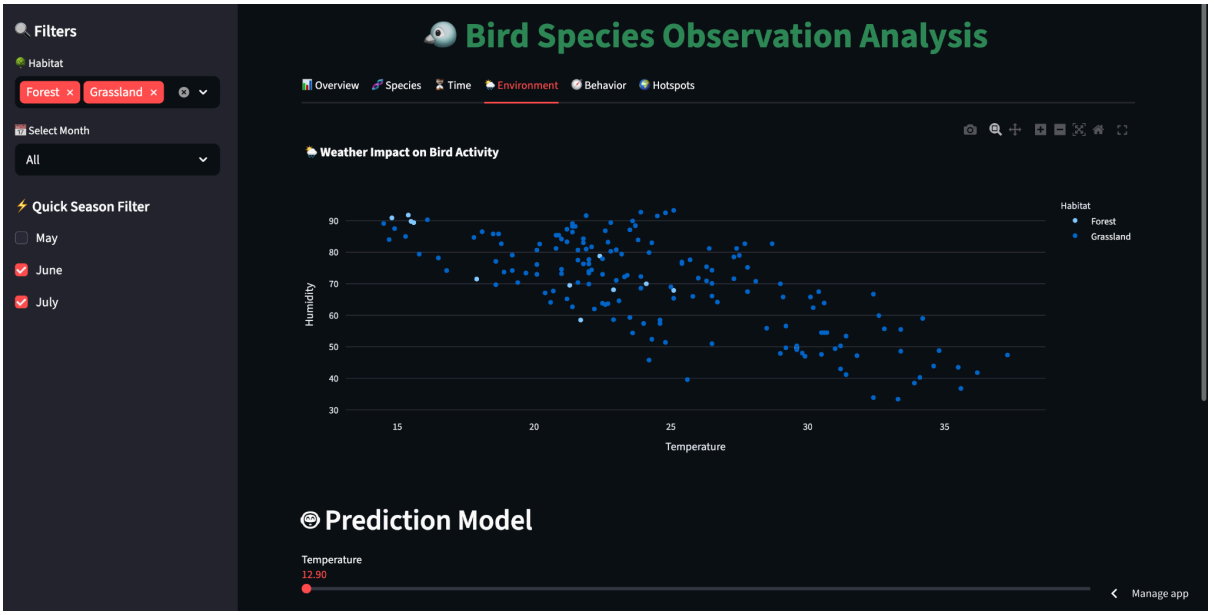
2. Species Analysis



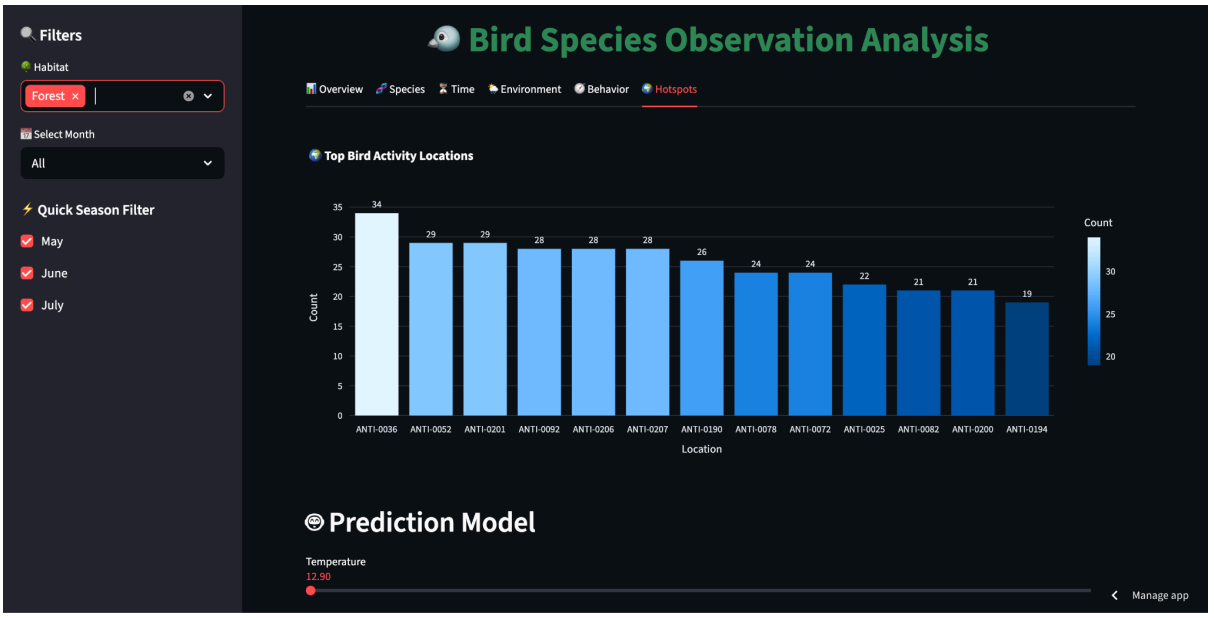
3. Monthly Trend



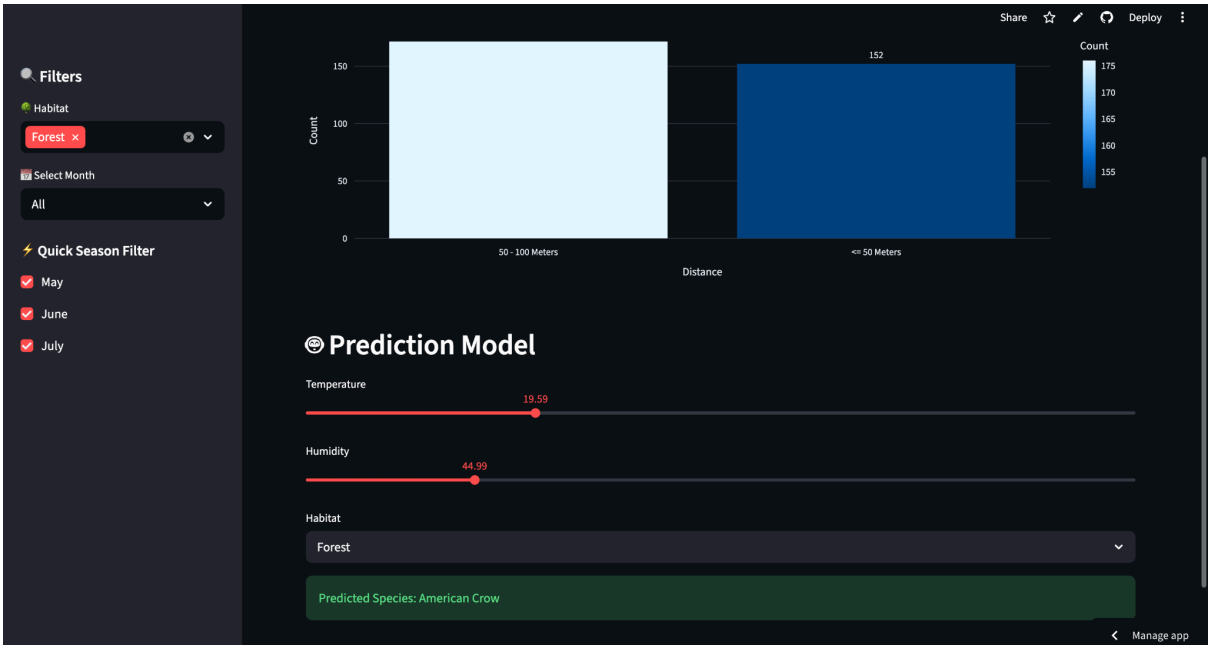
4. Environmental Impact



5. Hotspot Analysis



6. Prediction Model



🌐 Live App:
https://bird-species-observation-analysis-cqotyamp5rmtwacnpwmso9.streamlit.app/

💻 GitHub: https://github.com/AchintyaKrishna/Bird-Species-Observation-Analysis

🎥 Demo Video: https://youtu.be/rVKO2Cp0dlU